

Subtitles of PoS videos

https://www.youtube.com/watch?v=ulqv_tX_uAY&t=1s

hi

i'm here today to talk about the short
essay written by duncan watts
about why everything is not obvious
so it's it's kind of an overview of of
the paper and i'm going to talk about
the key questions that are asked
and the kind of answers that that duncan
watts provides here

um and and the

Paper Summary

four primary questions you could say
that he's asking in this essay here is
why do we often think that social
science is obvious
after the fact what is it about social
science it seems different
from quite a few of the other sciences
for instance physics or engineering
um you could talk to an every
layman on on on the street and and they
will have an opinion and they will think
that they know something about
unemployment or management or strategy
and then often results that we get might
seem obvious

and then sort of the point one of the
key points of this article is then to
argue and show

well it actually it's actually not
social sciences are not
as obvious as it might seem um they are
particular and are different in certain
ways and for instance

the natural sciences and just also short
indication of what's then the solution
what

how do we solve this particular
challenge here

so it's it's a really foundational
philosophy of science perspective that
also has implications for how we can
actually do science

and in particular question three but actually also so the overall essay very strongly relates to the nelson paper that that is also on the curriculum for for this class here okay so to Comparisons with Physics try to create a contrast sort of quite drastic and radical contrast to what social science is about let's try to think about a an example from physics that is at least to me quite mind-blowing so we have a guy called cavendish he worked in his own lab in england in the late 19th century and he built a device sort of this spec here more or less a wooden device but it had some some metal balls and some strings and things were attached to each other and basically what he was doing in his own basement in england in the 19th century was measuring the weight of the earth he was about one percent off it later turns out with modern science i think that's reasonably okay so again sitting in his basement being able to measure the weight of the earth and since we knew newton's laws of gravity he could also infer the weight of all the other planets and that's fair to say that this these results were not obvious at all if you talk to the regular person on the street they would not have been able to figure this out they probably wouldn't even have been able to figure out that you could measure this weight and you could infer these things and then you could ask yourself what is it that we social scientists or in

business administration what what can we
do with
in our lab or in our uh sitting in our
office trying to figure things out
it's certainly a very different field
that we're in
um and the sort of one point that is
often made that
should be obvious that social sciences
should actually be less hard than
physics or like literally that's the
expression that we use rocket science we
can
send rockets to the moon and and surely
that's more difficult than figuring out
human behavior the moon is pretty far
away
and that's sort of the puzzle we and
what is it about social science it seems
particular
Seemingly Obvious
let me try to get to this from from
another angle
um trying to illustrate this obviousness
so following the second world war
there was a really large-scale effort to
try to investigate
what it was like to be a soldier so they
were interviewing six hundred thousand
soldiers
um it's quite a large case study and one
of the key questions was who handles
life as a soldier better
and surely people from the countryside
they're used to physical exercise
they're used to being out in the open
maybe sleeping out in the open handling
different
materials and just living a quite a
rough life
and then whenever the main researcher
here he presented this story this was
also how he
told it and and people in the audience
would often go
surely you didn't have to interview 600
000 people to get to that result
but actually and that was a trick that

he often pulled with the audience it was
actually
people from the city it was it was
soldiers that had grown up in the city
that were
the best that could cope with being a
soldier the best
they're used to being in a uniform they
were used to working in a hierarchy
they were used to working at a specific
specific time schedule doing what others
tell them to do
um so this is a typical situation
in social science where both kinds of
answers completely opposite answers
could sort of seem obvious uh when we
are told about them
um here is just a slightly longer
explanation and a link to the story
about the
the american soldier and unobviousness
and we can think of other
examples here that from psychology that
it might
seem obvious that if you want someone to
say yes to a small request
giving a bad reason is surely better
worse than no reason at all
or if you want to be happier count as
many blessings as you have
count how many things that makes you
happy as possible
surely that's better than only counting
a few
we can think of other examples as well
right
surely but again um as as research has
shown
it's actually the case that giving a bad
reason
for in this case the research was done
on standing in a queue of getting to a
copying machine
and and and if you then gave a bogus
reason for why you caught in line
in front of the queue you said i need to
make copies well
surely everyone here in the queue needs

to do that it was a cue to the copying machine but actually it's just as good to say this bogus bad reason as it was to give a good reason because i'm in a rush and a slide here illustrates some of the other examples and one could come up with a range of other examples that that what might seem obvious is not necessarily so um but again this is a particular situation that we're in that that we if it it seems obvious and we should be but we are much better at rocket science a field that does not seem obvious we can send rockets to the moon or even people to the moon while figuring out what movie is going to be a hit or whether we should change the unemployment rate with how much or whether an advertisement actually works those are really really difficult questions and and these seems to be basically much harder than rocket science at least we're not making as much progress with them

Why It Appears Obvious

so question is and that's also what what's then is writing about here why is it that social science seems obvious what is it about this field that is quite particular and and for one is that it's it's it's familiar so we whenever we work for instance in business administration whatever bachelor project you'll write or master thesis you might write about how motivated people are or stock market or company strategy and these are all topics that we that we live we read about them in the newspaper we we think about them and we might have opinions about what is good management

how should a manager behave
surely this is the right way to do it we
don't in the same way live in a world of
atoms or black holes or quants or
whatever they are dealing with in
physics
so that that sort of one element that
can lead to
our signs being particular it's it's
familiar we face it every day
so surely we know something about it
but it's also related to human bias i'll
be talking more about in them
in a later lesson that we tend to after
rationalize after the fact
we tend to when we've been presented
with some information or solution to
something we tend to think
surely that's the way it was supposed to
be i could have told you that in advance
even though that we might not have been
able to do that
so this after rationalization also makes
things appear more obvious
after the fact and then i think also key
element that common sense
is usually quite effective in
everyday life
we get i mean it's quite efficient that
we know a lot about how the world works
or when we enter a room when we deal
with objects
we know a lot about we're quite
effective in in the way that we can cope
with everyday life
um on the other hand common sense also
has an answer to almost everything
birds of a flock flock together birds of
a feather sorry flock together but
opposites attract
it's bad to sweat the small stuff but
good to be detail-oriented and we can
sort of come up with
all sorts of common sense answers to
lots of life's
predicaments and challenges so
so what's here lists a number of
elements that

shows how social sciences and business administration in particular has a particular nature and can seem obvious because it's familiar because it is about stuff that we deal with on an everyday basis and makes us think we know what the social world is all about but and that's really also then one of

Why It Isn't Obvious

the key points
it's not it's not obvious predictions and explanations are really really hard in the social sciences because we live in such a complex world
when we try to explain whether a company strategy is going to be successful we have to take into account how the employees are going to develop it and how they're going to execute it and how competitors are going to react to it and whether there's a financial crisis going on at the moment and how customers are going to react and and etc etc they're all these different levels and they all influence and interact um which makes the social world extremely complex in a little sense of that world and we'll get back to that in a in a later lesson as well
but humans are also particular in the sense that they can change the behavior and change their minds if we figure out how the planets they evolve or go around the sun in the solar system i've got to change behavior tomorrow and they're going to continue in the same direction no matter what we figure out and what explanations we we create while human beings if you tell them about a certain advertisement strategy and why companies do what they do customers might react differently to while having that awareness um so so that also makes the social science world quite particular and

particularly hard
uh because human beings are not static
and just continue in
the absolute same direction as many
elements in a in a physical
world would do and then also
we often don't know the counterfactuals
we often don't know what
what could have happened but didn't um
taleb has a good example in his book the
black swan about
the greatest the person that could have
had
been the greatest hero and could have
had the greatest impact
would have been the one that had come up
with the law saying that all pilot doors
have to be bulletproof and locked which
could have enabled
avoiding the 911 terrorism incident in
the united states
um if there had been such a law but
there wasn't but what of all the
different
laws and rules and activities that have
been put in place that
that where we don't know what it
actually prevented
again we live in a very particular kind
of social world that it's complex human
beings can change the behavior
we don't know exactly what impact our
activities and actions have
um because we don't know what what could
have happened

Summary and Future Lessons

here a few more quotes from from the
paper on why social science is hard and
different from physics
so that was the the essence of the paper
that i want to present i wanted to focus
on well what are the key questions that
are being asked in this
quite short essay um
and i want to give this overview and
this understanding of social science i
think is quite
important for understanding the channels

that we face and
and the situation we are in in our
discipline
it has some consequences for how to do
science which i'll get back to in later
lessons
and then in particular i think it's also
a good thing to read this
paper because it relates so directly to
another paper nelson
that we're going to get back to in in a
later
lesson that so to further dive into what
is it about
social science why is it it's a
particular field
and and what philosophy of science
implications does it have for the
specific choices that we should make
when we're doing a bachelor project when
we're doing a research project or
whatever challenge we're we're facing

<https://youtu.be/EdCyxonpTxQ>

now it's time for an introduction to
what an experiment
actually is so
what is a randomized control trial
that's the same thing as an experiment
it's just
a randomized control trial it's sort of
the more technical term that makes
explicit the three different elements i
want to
get you as give you a sense of why i
experiment the gold standard in terms of
establishing causation
and later on in the live session we'll
get to internal or external validity
okay and and the aim or the learning
objective is that you will be able to
assess the quality of an experiment
whether it's in business administration
economics more in general or whether
it's in all sorts of other
fields and medicine or whatever whatever
it could be you'll

learn sort of the essence of what to pay
attention to in order
to be sure that this is an actual
experiment
Importance of Experiments
so this is the question that we also
grapple with in the form of video
how can we be certain that in fact is
due to one specific variable
how can we be certain that it wasn't
just a random correlation or that
there aren't other factors that we
haven't thought about
or the variables that we hadn't thought
about that actually are the key
that are key and as the chief economist
that
google said there in 2016 to find out
what happens when you change something
it's necessary to actually change it
that's really the essence of the
experimental approach
is that in order to be sure that
something has an effect
you need to try this effect out and see
what the results are and and i also
refer to someone from google just to
highlight that
experiments of course isn't just
something that we as researchers carry
out
um these numbers are even now a couple
of years old so they're probably
outdated and much higher now but at this
point in time
every year um google and amazon and
netflix etc ran
thousands and thousands of different
kinds of experiments so in that sense
you've also been exposed to a lot of
experiments just by being online
because netflix constantly tries out
different ways of presenting their
information to the consumer in order to
see
how do we get people to consume the most
um jeff bezos from amazon even says our
success at amazon is a function of how

many experiments we do per year per month per week per day so they do this all the time and again of course they also do it in a in a proper way there's no reason to sort of do and a half-baked experiment that doesn't work you have to do it in in the right way and another illustration of this comes from from this sort of small article here an experiment experience that ebay had a couple of years ago um basically they had the experience that the more they changed and ad expo the more they used ads on google for instance the bigger the change in the more they also sold their products so basically an ad in google is you you probably know this or you know all the other search engines if you type in a word there will often be an ad associated to it so the top link here is actually not sort of a and a link provided by your google search as such it's a sort of an additional ad feature that is on top of it that google is making money on and what ebay was experiencing is that friends every month or every quarter they were sort of increasing the number of ads they posted and the more um ads that were exposed to human searches well the more products they solved so fantastic their ads work right well it turns out they actually got sort of an unplanned way of testing this because in in relation to a negotiation they had with another company they wanted to show well how big was the effect of this this ad and so they shut down the ad feature and what effect was there

well okay this link says absolutely nothing that might be overdoing it a bit but as you can see visitors via the paid link versus visitors via the unpaid link the moment the ad feature stopped the number of visitors that went to the unpaid link that is clicked on this one which doesn't give google money it's only this one here that does just increased a lot so um so so basically what ebay and this is was at this point in time also a very large company run by very very qualified people but they for years did not realize that their ads did not give them the value that they thought it they would simply because they hadn't run an experiment to test it so just a sort of very sort of business oriented example of the need to run an experiment even in situations where you think you have a pretty good idea of what's going on because you could after all be wrong it might appear obvious as duncan wants to say but it wasn't after all so what do what

How to Conduct a RCT

do we do what do companies like amazon and google et cetera do when they actually run an experiment so the essence is as we often use the the medicine sort of a headache build example because it's it's quite intuitive you want to find out if a headache pill is effective so you find a number of people let's say 100 people that have a headache all have to have sort of the reasonable same kind of headache you then randomly divide them into groups one group gets the pill one group doesn't get a pill and in that way we can then see over time for instance six hours how many of the people in this group

that got a pill have a headache
versus how many in the group that didn't
get a pill it turns out that the
reason why i have three distinctions
here is that it turned out that there is
something called a placebo effect
just getting a pill makes you think that
there might be a positive effect so just
the interaction with the doctor or the
nurse or even the pharmacist
can have a positive effect so it turns
out the placebo effect is actually quite
substantial in terms of reducing
headache
um it does um
in fact have a substantial effect what
we're interested in
is the actual chemical effect of the
content in the headache pill
that's what we're really after we don't
want to sort of identify a placebo
effect
so this is the way we do it we have a
number of people we randomly allocate
these people into one group another it
should not be
i have the most headache therefore i
need to get into group x or my headache
isn't that bad or i don't like
pills so let me put it down here then we
could certainly see other kinds of
effects
that if we end up having the people with
the most headache getting pills
well that might also be most likely to
get their headache reduced by a pill
so we want these groups these three
groups here we want them to be
as absolutely similar as possible
and as indicated this little picture
here of course people shouldn't know
that they're getting a placebo pill or
that they don't get a pill or i mean
this is
ideally people don't know which
condition they're in because again if
they know that they're in a certain kind
of condition

they might adapt their behavior
accordingly
and only one variable should be changed
so
ideally these people in these three
different groups they behave completely
identical
just because you get a headache pill you
don't go out and play soccer and
get the football in your head and thus
increasing your risk of getting a
headache
i mean you should sort of have the same
kind of behavior in all groups
before and during and after the
experiment in order to get as clean a
test as possible
we'll see a bunch more example of this
as we go along this is just to
really identify the the essence of it
it's a randomized
of random allocation it's a controlled
trial so
we have to control that all these groups
are sort of
similar in as many ways as possible
and it's a trial or it's a randomized
control treatment there's only one thing
that is being done differently so that's
the weather
um rct it's not an rtc that's what the
rct is is coming from rct
and let's take another example that's
from the o'reilly text
in the curriculum so o'reilly wanted to
find out
if seeing the fruits of our labor makes
us more productive
so he did an experiment where he took
a bunch of students at harvard and
randomly allocated them into two groups
in one group
well in both groups they had to build
lego bionicle figures
but in one group the um the
figure was disassembled that was taken
apart
right after they had finished building

it so in this specific
uh group right after you've you've
finished building your your bionicle
someone is standing in front of you and
taking it apart it's very clear that
there's really no point to building it
in the other group what it's called a
meaningful group it actually just means
that no one took it apart
so as similar setup as possible
they both build lego bionicles they both
get the same amount of money
then the question is how many are they
willing to to build as you can see you
get three dollars for the first
270 for the next 240 to one etcetera so
for how little money are you willing to
build
the next bionicle and it turns out as
you can see
in the meaningful group they were
willing to build 11 in the
in the other only set so here we have
another randomized control trial where
we have
a group of people randomly allocated to
one group or another again they're not
being asked
so would you feel it annoying if someone
disassembled your
bionicle right in front of you and if
they say yes they get into the
meaningful group no
we don't want this kind of selection we
want a completely random allocation
of people into one or the other group
the condition should be entirely
similar this shouldn't be sort of be
happening in in sunshine while this
happens in rainfall
all other factors should be the same
only one variable should be changed
the fact whether it's disassembled or
not
and again of course we have to remember
this is an average causal effect on
average
they built 11 and 7 here of course there

was a distribution of
of number of bionicles they built across
individuals
so just sort of to really reiterate all
these elements here randomized control
trial
we have an allocation to each treatment
so we have a group of people
that we then randomly allocate to each
group notice that the people in each
group they're not necessarily identical
i mean one person can only get into one
group you can't
split and copy one person and then make
sure that
they're completely 100 identical we
can't clone people
organizations or whatever we're starting
like that but we can assume that if we
had a group of people
and we then randomly allocated for
instance via coinflip into one of the
other group
they are as similar as possible they're
not going to be identical
but they are as similar as possible one
group plays
it has a one kind of bionicles condition
the other one the other one
and we can then see the results but the
idea is that if you
because you have this random allocation
to each group then these groups are as
similar as possible um
and for instance you can think of it in
terms of gender let's say you have 100
people and you divide it up into
by by a coin flip into one of the other
group if you start with 15 men and 50
women
most likely you have about 23 24 25 26
or something of each in each group
it's not going to be completely
identical number
in terms of gender but it's gonna be
very very similar just because of
of how randomness works when it
accumulates

here i have a picture of two classrooms
because we get to a classroom example a
little bit later
but just to show that if you run an
experiment in a room or
wherever you run your experiment it
should be completely identical
um the setting should be as identical as
possible
and then which then enables there to be
only one treatment there is only one
thing that is
different from one setting to the other
What Is NOT a RCT
so let's try to
think about this a bit um
and and think of this example here it's
very famous example it was also
documented and showcased on danish tv
it's the marshmallow example
so it's an old study from from back in i
think the 70s
where four-year-olds were offered one
marshmallow now or two marshmallows 15
minutes later
so as you can see there's a kid sitting
in the room there's a marshmallow in
front of them and they can eat it now or
they can wait 15 minutes and they would
get two
this is a randomized control trial you
can if you want to just pause the video
and reflect on it
and consider whether it's sort of liv it
fulfills the criteria we have oops there
was one boy who couldn't wait
um who knows maybe they
weren't telling the truth when they were
promising two marshmallows later so
better eat it now right
um in this situation
yeah better hide the plate then no one
knows you ate it um in this situation
it's not a randomized controlled trial
because there is no sort of
everyone is getting the same treatment
everyone could get two
marshmallows later so they're sort of

self-selecting into
either eating right away the one or
getting to getting an additional one
later on
there was not a random allocation in
terms of um
of who gets how many marshmallows so
it's an exact identical setup and people
sort of self-select into
whatever outcome they end up with you
could have done an experiment by saying
so what is the impact of having to wait
seven and a half minutes versus 15
minutes that would have been
saying a direct comparison let's take
the exact same setup
put kids into a room some kids wait
seven and a half minutes other kids wait
15 minutes
let's do a comparison of how many
marshmallows are being eaten right away
that would that would have been a way to
say it's an experiment so we need to
make sure there is a randomization
element
that's what this example is is
highlighting
um sort of the randomization element we
need to randomize people into different
settings so there can be a variance of
treatment so again also the
treatment is not identical there is not
sort of a thing that is different
across the situations okay so let's
take another example and read about this
basically a
reference to the bible and what could be
the first documented randomized control
trial if it actually is one
so again you can pause the video and
read the snippet that i've inserted
there and think about it
um i'll go through it here so the
essence that
the way that is being described in the
by in the bible is
daniel is a fellow who doesn't want to
eat meat and insist that it's healthier

not to eat meat
king was interested in this okay how
about
we test your servants for 10 days um
or daniel's is proposing this so we
should test your servants test the
servants for 10 days
give us nothing but vegetables to eat
and water to drink and then we compare
our parents with that of other young men
who eat the royal food
and then to see what's going on so the
king ends up comparing the four non-meat
eaters including daniel
with other meat eaters and then after 10
days
after this 10-day treatment they are
then compared and then
to assess are those at a meat do they
look healthier than those that did not
eat meat
and and and here's there seems to be a
treatment
meat or not but
concerning the allocation of people into
each group
it then there is self-selection in the
sense that daniel who came up with the
proposal
and who doesn't eat meat he is himself
in the non-meat eat
group so so maybe daniel is not eating
meat because he
already is healthy and can sort of um
get by by not eating meat so maybe sort
of the effect is coming from somewhere
else in that sense that
the two groups are contaminated so to
speak because there is not a complete
random application from one
in in one group and the other group and
it could be the case
that um the people that ended up in the
non-meat eating groups they were already
healthier in advance
they were already maybe the kind of
people who sort of could
eat enough vegetables and still look

healthy so
we need this random allocation and
that's why this is not an actual
experiment even though it's pretty close
and pretty impressive that
according to the bible this happened at
one point
okay let's do one more example um
researchers interested in the impact of
rewards on behavior
so in one group 50 people are said to do
a task as many times as possible and
receive one crown per task
the other group 50 people are to do the
same task as many times as possible and
are told they will receive
a 1.50 kroner per task and the goal is
to solve 50 of them
pause the video if you want to reflect
on it maybe you already are at the
answer
two things are going on here we are
comparing one chrono per task with one
and a half chrono per task
but on top of that one of the groups
we're told that there is a goal
involved you should solve 50 tasks goal
setting behavior
does have an impact as you know from
organizational behavior so
the groups are not completely identical
this suggestion should also have been
given to the original group
in order to compare the two settings
and of course those in one group
shouldn't all
shouldn't know what is happening in the
other group because you might be annoyed
about
landing in the one crown of a task if
you could have landed at one and a half
and that was just bad luck
and this annoyance could then lead you
to performing more poorly or what sort
of whatever i think that could be there
so
as highlighted here we should not we
should never really know what is a

treatment in other groups so if you participate in the in the kobe lab in the in our social science lab here at old university at bss that's why you should not sort of know what treatment you're getting or what treatments others are getting because that could lead to some kind of hawthorne effect where you know you're being observed on this particular variable and and maybe you'll adapt your behavior according to it it's a challenge of lab experiments that we'll get back to

Key Elements of RCT

so key elements that should be a randomization process whatever group of people you have you should allocate them randomly into one or the other group next time we'll get back through in the next video we'll get to thinking about in the o'reilly experiment with bionicles it was actually only male students from howard that participated so is that then still completing the criteria of randomization second point is you should manipulate the treatment should have been only one variable comparison with control and treatment there should only be one thing that is different and then it should be a control setting so everything else should be the same so you should really make sure that if you if you have two different groups at two different times that you try to make them as similar as possible of course nothing can be completely similar if you run one part of the experiment at 10 o'clock and the other one at 11 o'clock well people at 11 will have gotten out of bed a bit later and maybe the sun is shining a little bit more although the room is lighter

i mean it should basically and in essence be the same but of course even the air that is being breathed in the room is not completely identical you can never expect sort of different groups to be absolutely completely identical what you're aiming for is that it should be as identical as absolutely possible this was the essence of a randomized controlled trial we'll get a bunch more examples and illustrations and also of different kinds of experiments in the coming videos and sessions but these were the basics as they are

https://youtu.be/Wz-1b_B6InU

n this video i will talk about correlation versus causation these are concepts that i'm sure you've heard before i'll try to dive a bit deeper into them here in this in this video so idea is to get a sense of what do these concepts mean and how and how are they different um but also talking about what does it actually mean for something to cause something else and should we even think of single or multiple causes and what is a cause in the end the meaning of the phrase is that we often claim that something causes something else so the meaning can be sort of instantiated in many different ways we can say something leads to or brings about or harms impact reduces something increases something else or decreases something so we haven't we use it often in everyday language and say that something causes something else and this is what we'll want to to get a sense of and this will then lead into part two of this lecture

where we will dive into the golden standard for establishing causality which is experiments or randomized control trials the same thing

Correlation Fallacies

okay so we have this idea and i think we have this notion that we know that correlation does not equal causation just because some things are correlated does not mean that they are sort of causally linked to each other

so this title mission page has a bunch of examples of weird stuff that is just extremely highly correlated so the consumption of margarine is 99 correlated with divorce rates in maine and it's not like people get divorced because they consume too much margarine or the other way around

um there are a bunch of other examples it's just because there's so much data in the world that in the end a bunch of stuff will be linked

what we want to get a sense of is when can we be certain that there is a causal relation because what is going what is happening in the human mind and we'll get back to this in lesson eight when we talk about cardamon's text is that we as human beings have a tendency to

to see stuff being linked in a causal way that's how we make sense of the world

we link things and that's why we sometimes end up linking things that shouldn't be linked because we have this natural tendency of of linking two different things together

in this case with margarine and divorce rates it's probably so obvious that we should be too worried about it but there are others more sophisticated and more subtle situations than we'll try to address

where this is not so straightforward but

again we are after the
cause a causal effect so sun can
cause more consumption of ice cream it
can also cause people to get sunburned
but of course sunburn doesn't cause more
ice cream to be eaten or
ice cream an increase in the consumption
of ice cream doesn't cause more sunburns
but these things are correlated the more
ice cream that is being eaten in denmark
it's correlated with how um how much of
a sunburn that individuals are getting
but again this is because of another
causal factor and that's really a
key thing that we'll try to get a sense
of here is that
we need to be sure in the way that we
design our research
that we can be certain that it's the
actual causal factor that we've gotten a
hold of
and not some more sort of spurious
correlation whether something is
correlated but
it's because of another factor and you
can maybe put the video on pause and
just read this reread this piece of text
that's from the bowerman
from the curriculum i want to sort of
highlight the essence of it because
it illustrates how we can sometimes
identify something that
clearly seems to be related to the
but but actually might be sort of uh
hiding another underlying causal factor
okay so this is a very famous example of
a psychologist government
who interviewed married couples
and based on these interviews based on
these conversations he had with these
married couples
he apparently had a 94 success rate in
terms of predicting whether these
couples would get divorced within three
years
so calling it a success rate and a
prediction might be sort of a little bit
off

so but he was very good based on these conversations because he could identify how sort of how much content and criticism and how defensive they were when they talk to each other these married couples he could then use that as a sort of information to get to his prediction but this is then often as baromen points out often misinterpreted because it's not like you could then divorce proof your marriage and change the way you communicate and then make sure you don't get divorced because the issue is that these factors of communication are not necessarily the causal factors that create the divorce it could often be the case that it was actually another underlying factor whatever that could be of disagreement or or unhappiness that caused the poor communication and also caused the divorce it's basically the same thing as we saw with the ice cream example that these two are strongly correlated but in the end there is another causal factor that is causing both of them so the point here is that um we can't be sure that the outcome is due to these communicative behaviors and we certainly can't be sure that changing these communicative behaviors would actually change the outcome and it turns out that gottman actually wasn't that successful in sort of making sure that people didn't get divorced um so again what we are after is that we want to create a research design where we sure that we've identified actual causal effects and not what we can call a confounder confounder is with something sort of

interacting or
mixing or mixing something up and in
this case it could be
that the communicative patterns
are not the actual underlying core cause
but they sort of
confound they confuse our interpretation
of what is actually going on
so this is something that we want to
learn using
uh our insights into research design
that will be generating
to become better at making sure that
there is this causal effect that we have
another challenge that we already heard
about in lecture one was is this thing
about selection we have to be sure that
what we're seeing as
a sort of a causal effect it's not just
because we see
a biased um selection of data it's not
because we only see
part of the data we only look at
successful entrepreneurs or only look at
the planes that actually get home
um in the second world war that then
leads us to a certain kind of conclusion
for instance of how to become a
successful entrepreneur
so that can often then also be an issue
that um it's because we only look at a
subset of the data that we can end up
thinking something is causally related
even though that it's not we might end
up concluding that all successful
entrepreneurs
sleep at least five hours per night but
but the unsuccessful entrepreneurs
probably also sleep five hours per night
so it's not really an actual causal
relation
in a meaningful sense and the reason why
The Role of Media
all these things about causality is
is really important is highlighted in
this slide
which i'll try to really go through in
detail because it's quite important

the reason is that scientist and especially the media has a tendency to oversell a certain story okay so what are we looking at here on sort of this you could call it the x axis from left to right we have a categorization of the strings of causal interference how certain can we be that what we've identified is not just a correlation but it's actually a causal relation as you can see this is not sort of either it's true or not it's sort of continuum to the left there are lots of problems with the research design and we can't be certain that this is actually causality the more we to the right we can say okay this is very very likely that we've identified a causal effect what we see when we look at the academic studies and this is then the group of researchers who've looked at the 50 studies published in this journal plus one that have been shared the most on social media so they found the 50 studies that have been shared in total 1.3 million times so these have really been shared a lot these have been read by many people um i've looked at these 50 studies and then a group of of of experts have rated these studies and assessed well what's the strengths of the causal interferences that these studies could make as you can see only four percent only two of them were really in this very high category one was in the higher category but a big bulk of them were in sort of the low or the moderate categories um meaning there were certain kind of problems in the research design that meant that we can't be 100

sure that identified an actual causal effect
if we look at the studies and themselves well how what what cause language do they use in these 50 plus one article so what language does the authors themselves use this the first row
is outside experts that have re-categorized what this research design and the studies actually could do the next layer here is the researchers themselves who've authored these 50 studies how they categorize it and as you can see we've suddenly moved from four percent than this very high causal we have identified a causal effect to 20 so five times as many are certain that they've actually identified a causal effect and as you can see they are more certain about causality than than the outside experts okay so who who knows maybe the authors are actually right this is also to some degree subjective assessment but when we then see and this is really the key point when we then see the media studies suddenly 44 of all media sort of like newspaper articles newspaper articles about these 50 papers almost half of the used language is caused by is due to more coffee drinking leads to higher whatever or eating more chocolate leads to higher whatever it could be they have identified they they claim that they sort of portrayed as a causal effect even though the original studies actually didn't try to do it themselves or certainly shouldn't have done it themselves so it's a fundamental challenge that we see

both in scientific studies but in particular in the media stories in newspaper articles about scientific studies that they tend to oversell how certain we can be about the conclusion in this in this research and and this is then also why i think it's a fundamentally important competence to have to be able to read a newspaper article and say well not just be skeptical we can all be skeptical but be skeptical in an informed way where we can say well because the data has been collected in this way i actually can figure out that we should not trust this particular conclusion because for various x y that reason whatever happens in the particular context we cannot be certain of it and we'll see a lot of examples of that both in the next lesson on experiments and also research design in general we'll be getting the tools to become much better at identifying how much we should trust the results of a given study and this is even if we don't know the theory that it's involved we don't have to be field experts there are some sort of fundamental things about research design that we can pay attention to so that's sort of the challenge this is basically just to identify the challenge that we have um and showing that this is a challenging newspaper articles and what we want to get is the tools to be better at this

The Nature of Causation

but i also want to sort of say a few more words about what what does this causation then mean okay we know that it means enhance or impact or leads to or has the consequence so all these other sort of

everyday terms we can use but but is it actually
the true causal effect and i've sort of already indicated that
no i mean this tweet really highlights it or summarizes it
causality is not binary we should not think of causality as either being there or not our causal explanation can have various kinds of certainty so it's continuum
like we see here it's a continuum of being very uncertain to being very certain about the truth
of a causal effect um
and and so while we in theory of gravity could say well it's really deterministic and in our field it's very much probabilistic
so the impact of incentives on solving simple tasks
is fairly high but it's certainly not a one-to-one match
so that's a key thing in social sciences we should think of
causality as being likely
in addition to that we should also think of well
not it's not only a question of whether it's sort of a one-to-one deterministic match or whether it's just probabilities but should we maybe often think of multiple causal effects
because the word cause might make us think of that there is just one thing one thing causes something else
sunshine causes sunburn but as even john stuart mill said quite a while ago
it's rare that there is a single antecedent there is rare that there's a single thing that comes before
the the consequence there's usually a sum of several
things that need to be in place for an effect to happen
so often we shouldn't think of one cause
yes smoking has a big impact on lung

cancer but there are a number of other factors that can either decrease or even completely eliminate the potential impact of smoking even such a simple question as does lightning cause fire actually illustrate this because yes i guess in some sense it does you can see situation where there is a lightning strike and a fire emerges but i mean is the dry ground what's the temperature has it rained are there other random factors that play a role it's not sort of lightning as a single factor that will always sort of be the sole reason for why something is likely to why a fire is might emerge um there are often multiple causes for something to happen an organizational success is not depending on one single factor is depending on a long long range of various different factors and that's also already emphasized it's not just a matter of probabilities it's not just a matter of that that the causes are often coming from multiple places it's also an average causal effect that we usually dealing with in social science so just because there are some humans that don't respond to a cause in the same way as others i mean that's just the way life is and then something will also dive deeper into we still think that there generally is a causal effect even though not everyone benefits from a certain kind of drug as the medicine slide illustrated the last time and then also just it's a very basic philosophical question of whether causal effects are even real

because we can't really see them um
well we might be able to see this feel
the sunshine that could sort of create
the sunburn
but it's not like we can really dig down
into the skin
just with our bare eyes and see the
actual causal effect
of how financial incentives influences
human behavior again
we know what a financial incentive is
and we can look at the performance but
there's something in between here that
it is sort of difficult to
completely specify what it is this
causation
and it's something that we'll get back
to in in several other lessons
um to what degree causation is a feature
of the physical world
or just a convenient way to think about
relationships

The Aim of Research

so overall and that's really also the
key message here is
yes coalition does not imply causation
sorry
coalition does not imply causation um
but it's evidence for causation so so we
should sort of be cautious about this
it's often said correlation does not
imply correlation as if
correlation does not imply causation as
if i mean we don't have any evidence for
it well they would imply is really meant
to mean something strong here because
correlation certainly
indicates causation as also written in
the baromen
article it doesn't imply causation but
it does waggle its eyebrows suggestively
and gesture effectively will
mouthing look over there so if there is
a correlation it's certainly worth
looking into
maybe we find out that there wasn't any
relation but but it's
certainly something to to pay attention

to so we shouldn't really
say doesn't imply causation to make it
sound like there's no relation at all
if there is a correlation there's a good
chance that there might be some cause of
the factor associated with it
but the aim is to reduce uncertainty the
aim is to figure out how can more
certain about the conclusions that we
come up with in science
so science operates as a procedure of
uncertainty reduction as we'll hear more
about in lesson 10.

so it's not truth in the sort of
complete capital t kind of way but but
something similar to it that we often
have to settle with that we get to

How to Infer Causation

but the question then still is how can
we be sure that
actually at x actually causes y uh let's
take an example let's assume that out of
a

thousand firms 500 have decided to
increase salary in 17 and
the other 500 did not firms where salary
was increased they actually had a higher
productivity in 18. let's say this is
how the correlation looks like
can we be certain that the salary
increase

actually led to this higher productivity
well these apparently the 500 firms in
this hypothetical example just decided
to increase their salary

but but why have these particular
companies increase the salary why
have we sort of why are these have the
these firms selected into the salary
increase maybe because they knew they
were going to do well in 2018

and they want to be sure to to keep
their employees and they could afford to
keep their employees

so maybe it's actually just because they
knew they were going to do well that
they decided to preemptively raise the
salary to keep their key employees

maybe it's just better economic conditions in a particular industry um that that sort of have led these two to a higher demand and then could uh facilitate higher productivity or maybe it's not the salary increase maybe it's something else

recognition or new leadership that influence behavior these particular 500 firms so maybe it's the new leaders that decided to increase salary that then also decided to do something else so

the thing is that with many kinds of situations like these where it really appears to be

i mean it would be meaningful if the salary increase led to higher productivity

but again we want to be really sure that it was this particular factor

and not some other factor that we're not just paying attention to

so how can we be really sure it's the randomized control trial it provides what we call a counterfactual that betterment talks about in page 11 and 12.

so we have the medicine example we have a group of patients

some receive the drug and some don't receive the drug in this way we can not only say

let's take a hundred people we give them all a headache pill and see how many don't have headache afterwards that could lead to the conclusion that that headache pill is really really effective because over time almost everyone loses their headache

but but what we want to do is to compare in a given period of time

50 people that get the neck pill and 50 people that don't we want the counterfactual

we want to know what would have happened to group b

if group b had received group a's

treatment

um and this is then what we then want to look into with the randomized control trial

we want to find out how the randomized control trial how an experiment can give of this almost certainty that it's the headache pill it's the intervention it's the salary increase or whatever the example is that led to the causal effect so how is it that we should design such a trial in order to be sure about causality

Summary and Next Lesson

so that's the aim of of of the next session

um is to get this understanding of randomized control trials

the aim of this one was to say look let's get a good sense of the concept of correlation causation

what are some of the challenges of why sometimes something might appear to be a causally related

why might it because there are other underlying factors or because of selection effects

and and we also then went through well what does it mean

for something to um what does it mean to say that something is a cause and effect should be think of as single or multi-causes

um and and and

and in social science we should sort of settle with probabilistic causes that have an average effect

sort of thing some general insights into what it means when we're looking for causality and

this then we're getting some insight into how to actually do that in the next videos and sessions

<https://youtu.be/TY1xoPon42U>

we will now continue and focus on

experiments
but while we in the former session focus
on lab experiments to illustrate
the nature of a randomized control trial
we'll now
focus on what we call field experiments
and
and i'll try to explain what these are
why we
run them so what are the advantages
and disadvantages of running a field
experiment compared to a lab setting
Synopsis of RCTs
so again just sort of to reiterate the
main points
the randomized control trial is a
particularly
strong and useful way of trying to get
to the
causal nature of a possible relation
between two
two things so because we
randomize participants into two
more or less identical group and make
sure the environment around the
experiment both in the control group and
the treatment group is as identical as
possible
and we only change one thing then we can
be quite confident
that we have established whether
something works let's say it's a
headache pill
if we gave the 100 give the headache
pill to all hundred people
and and let's say they all had their
headache removed well
we we like the counterfactual we lack
insight into well what would have
happened
if some of these people did not receive
a headache pill and that's what
randomization is for that's why we
carry out an rct but we also have to
Internal and External Validity
remember as
we heard before that causality should
not be considered as a binary thing

and the validity of an experiment is not
a binary
thing it's not yes or no or false
it's more of a continuum
and this relates both to the internal
and external validity
internal validity being what is the
truthfulness of what is the validity of
this particular
study in this particular context um so
of the riley study with bionicles
how certain can be that we can trust
these particular results with these
particular groups of people in this
particular setting etc
while external validity is then saying
well
we are not only interested in figuring
out
whether this one particular study and
with
but it's all it's with its particular
setup
is true we are actually also interested
to know can we learn something about
the world more broadly um so we
we focus quite a bit on that on friends
used o'reilly's bionicle
study to to make us think about well
what's the strengths of the external
validity
to what degree do we think we can say
well it happened to have been run on
male harvard students
do we think that this also applies to
the population more in general
the study relied on a very very simple
task of building bionicles
do we think it will also be meaningful
in in other settings where we do other
tasks
um so we we
the aim was to really get a sense of
that we should not say a study has good
or bad external validity we should be
more specific
in terms of thinking about interaction
between selection and treatment

interaction between setting and
treatment
it's run quite a few years ago would the
results still apply
into the social world we see today etc
so these were the
internal and external validity
trade-offs that we saw in
lab experiments and
and and and and it was clear that
especially external validity
was challenged i want to sort of
give a little tweak to the concept of
external validity because external
validity is a very broad concept and
while ecological validity that some
people also sometimes use
is more literally whether whatever
study you've done it also works in the
real world and
and here we have an example i've
mentioned my studies before
we actually have a study that shows that
if you run
studies that you usually run in a lab
with lab mice
and then put these mice out into the
natural habitat it turns out that some
of the results we get are actually
different
so it matters whether we have people in
the lab or whether we have people in the
real world or
here it was mice we had in the lab or
the real world so that's what ecological
is saying is how well are the findings
applicable
to the real world and everyday life so
this is a subset of external validity
but external validity is completely
broad it can be generalized to other
settings
another lab in another country other
kind of participants
this is whether the bionicle study would
also make sense
in an organizational context for
instance

and the point being that you had to
build bionicle that might be dismantled
right in front of you
for very small monetary rewards does
that really match
what we think happens in organizations
do they really carry out such small and
simple tasks
are things really dismantled in this way
so that's the that's the challenge of
lab experiments and that's why we
um want to focus on well well how can we
then get
stuff how can we get these studies out
into the
real world and that's really the promise
of field experiments
because they are run in the field so
sort of to to to highlight the point and
put it slightly differently
so that even if a lab experiment
is as well done as well as possible with
as high internal validity and external
validity as we can sort of imagine
um it's still often generally based on
student samples it
still will take place in artificial
environment people know they're part of
an experiment when you you
sign up to be part of something you you
show up
and you get paid usually these
experiments are often also fairly brief
they might last a couple of hours if
they're really long they might last half
a day
um so what if we want to study real
world situations over a long period of
time and
really involving real estate where like
whether you get fired or promoted or not
as an employee in a company for instance
rather than difference between
getting 2.40 or 2.1 dollars for building
a bionicle we we want to get out into
Field Experiments and Examples
the real world and that's
what field experiments do we still test

claims of causal relationships we still
randomly assign subjects it's still a
randomized control trial
but it's distinguishing characteristic
of field experiment is that conducted in
real world setting
and often unobtrusively meaning that
participants might not know they're part
of an experiment i'll show a couple of
examples of that
here in the following slides so but the
key thing is
they're done in the field i mean the
name sort of gives it away
um just to give an example of it so
a riley's bionicle study was trying to
figure out
whether it's important to see the fruits
of your labor and whether that makes you
more productive that was a
o'reilly sort of setup that's his claim
that's what he wants to investigate
and the argument is that even sort of
such a silly situation where your
bionicle is dismantling right in front
of you
um this is sort of counterproductive but
it's and and and it's one thing that
it's done on males only
that are really good in principle sort
of have dealt with that
but can we actually in a meaningful way
translate that to into a proper
organizational real world setting
well that's then what adam grant is
talking about here and
and and the study that i'm referring to
here on the right
is well we could answer the same
research question
so roughly the same research question by
saying i want to investigate
how important meaningful work
experiences are
and so here is an experiment done on a
group of employees into the marketing
company that were to call people
in order to raise funds for for a good

cause of sorts
and and what they've done is they've
created four different conditions
an experiment doesn't have to be only
two or three conditions it can be as
many as you want almost you're still
sort of
comparing across these conditions so
here we had a control group
here we had a group where the employees
at the telemarketing company
they they sort of heard about
good stories that happened when they had
actually raised funds for the cause that
they were arguing for
this did not really make a big
difference
here in this condition they actually
heard about life-changing stories let's
say they
managed to raise funds for someone's
education and they heard about the story
of
this person actually getting funds for
education and this was a big
deal for this person of course and this
had an impact on how much money was
raised this is what the bars here
indicate
how much money is raised by these
employees on average
yet meeting the recipients so actually
meeting someone who got
money uh based on the work done by these
telemarketing employees
really had a substantial impact plus 400
percent
in in in performance this really had a
substantial impact
so this is just to show a very simple
example the research question was
whether it's important to see the fruits
of your labor
well this is basically what adam grant
did in the study here
he compared to the marketing employees
that were either just sort of
in abstract learning about how important

their work was maybe they
actually heard about specific story
about any person who benefited from it
or they really met the recipients and
apparently this had a substantial effect
um so now we've investigated in a real
world context an actual organization
then we're divided into different where
the employees were divided into
different conditions
um and so the external validity is
clearly a lot higher i mean still
it might only work for tiller marketing
companies or maybe it only works in
i assume this was united states and
there can still be other limits to it
but we've taken it outside the lab so
the ecological validity is clearly a lot
higher
um in our our sort of
trust in whether we can generalize this
should be a lot stronger
and just to give you a couple of more
examples to highlight why we sometimes
need a field experiment to study
something so this is a very
well economic study big question is does
it actually
matter to to be sort of to learn about
management practices
so there were a group of researchers who
went to india
and did a field experiment on a large
indian textile firms
so they provided free consulting on
management practices
to randomly chosen treatment plans so
i would say there was a company textile
companies
who had various manufacturing plans
where they where they produced these
this is this textile stuff they produced
and
some of these plants which were selected
and they got free consulting so the
manager of these
these manufacturing companies they got
free consulting and then the aim was to

compare the performance of those sector
got the treatment with those that didn't
get the treatment to see well did
performance
increase and actually um productivity
increased by 17
in the first year
and which is quite i mean that that's
really a lot so these
facilities were quite big so this was
really really a big impact
but of course the question of this
management matter in principle you could
have created an artificial environment
and taught some
participants in the lab some management
practices and they maybe had to
create an artificial company and see who
sort of manages the best
it's of course far far more convincing
to be able to go
to two companies and
and then not just sort of do an
artificial thing but really do it on the
real thing
still in terms of external literacy who
knows if this translates the transfers
to denmark united states or china
we could sort of have a discussion on
that but it's it's certainly
um it's certainly a much higher validity
than any lab study could have
and actually the same group of authors
actually followed up
on these textile companies 10 years
later so they revisited the plans
in order to find out well do the effect
actually still hold here 10 years later
and as i show here there was still a
large and significant gap in practices
and even at large there was even a
significant performance gap between
treatment and control plans
so even 10 years later um
they could still see the effects of the
management practices they helped to
implement a decade earlier
so again this is then showing not only

does it have a short-term effect
but we can have a field experiment where
we can show that it actually also has a
very long-term effect
logistically this is of course really
difficult to pull off this research
because it literally takes 10 years to
to run but it ends up being a very
convincing and compelling case for
the importance of management practices
Is this a RCT?

okay i suggest now that you pause the
video and and read
the um the example given here it's taken
from
from the textbook chapter on experiments
so try to read this and consider is this
a proper randomized controlled trial
and what kind of experiment is this
so if you pause the video you have a
chance to read it for yourself
i'll just continue here so
key things to pay attention to is here
okay we it's
it's it's experimental participants are
sales people working for a
north american insurance company so
these are actual employees in an actual
company
and they were invited to participate in
a study
and they were only you had to be able to
commit yourself to four weekly training
sessions etc so
there was some kind of selection going
on here and 60 individuals volunteered
but then they were assigned randomly to
experimental and control groups
so it's a randomized if there's a
randomization going on maybe the
original 60 people
are not typical sales people so maybe
there are sort of particular individuals
who have the time and the
motivation to participate in some kind
of
management in some kind of training
program

but they were randomly assigned to one or the other okay then members of this the experimental group received training and self-management over several weeks and the control group did not receive it until much later in the future er effect a year later so that's quite a bit later but here we have an opportunity to then compare the performance of people that got the experimental treatment with the control group that have not gotten the experimental treatment yet and they assessed effectiveness at 3 6 9 12 months and it turns out the hypothesis was confirmed the experimental group performed much better than the control group at each point on the various effectiveness criteria employed um furthermore after 12 months when the control group went through the the training program the differences disappeared and they were sort of at the same level of performance so what's the takeaway here then we can do a randomized control trial in the field we can have a randomized setup we can sort of try to make a manipulation that is sort of quite clear either you get the management program or not but you could also speculate well how controlled is this environment how certain can we be that people from the management so those that were in the experimental condition did not talk to people that were in the control group we this is difficult to control we can't sort of um at least not in in most countries and companies you can't force employees not to talk to certain other employees and even if you did that that would sort of

create a weird awkward situation
so we cannot be 100 certain that there
wasn't some kind
of communication so we could actually
imagine that the insights from this
management
course sorry this training program was
quite
easy to convey to others and this could
have led to a situation where those that
did not participate in it yet
heard about it from those that did
participate and and
and then they all sort of improved just
as much so
if the result had been a null result if
they didn't find any significant
differences
let's assume that let's assume they did
not they were not able to find
any significant difference in the in the
experimental treatment and the control
treatment
well this could have been because the
sort of information was leaking between
these two different groups
that's um that's a challenge you have in
these field experiments so this way this
is a nice example of
the quality and the strength of a field
experiment you are actually finding out
whether this works for this particular
company but
but there are risks involved in terms of
not being 100 sure about the internal
validity
Internal Validity Compromises (Face mask)
another example that we also had the
last time was the weather face mask
protects against covet
the danish study that i talked about
last time and
just want to point out that's the field
experiment because it's run
in the field it's people that live their
everyday life
that that you either wear masks or don't
wear masks

so it's an experiment that is in the field and that's why we call it a field experiment we could question the internal validity because actually they weren't investigating whether face mask protects they were sort of more indirectly investigating whether informing about face mask and sending out face mask has an effect um as we discussed the last time but it's it's it's another example of how it's more difficult to control this if we had put people in a lab and just sort of hypothetically had had a group of 50 people where they were all wearing masks and another group of 50 people that didn't wear masks and there was an infected person in both settings then we could have investigated whether they got the disease or not this would not have been we would not have gotten ethical approval for running such a study of course um but in principle the internal validity of that study that sort of that mass study would have been better the internal validity would have been better but we want to investigate how people behave in everyday life i mean it's difficult to put people in a group where they might realize they were sort of in a situation where kobe plays a role and then they might change and adapt their behavior to that here we we had people as they were living their lives

Is This a Field Experiment? (Super Bowl)

an example i actually used last year as you can see the tweet is from february 19 so this is not covet related um you can pause the video if you want to sort of read

the essence of the study as it explained here
what it's being shown here is that influenza mortality in cities that participate in the super bowl so that's american football in cities that participate in the super bowl is higher
so what they're able to show is based on data from 74 209
they are comparing cities that end up having their team so to speak in the super bowl and then they're looking at how many the flu death do we actually experience
right after that super bowl is played in in late january or february compared to cities that where the team is not in the super bowl and it turns out that there seems to be an 80 percent increase in the flu death for the population over 65.
is this in a field experiment no and the reason is that the researchers have not allocated cities to participating in the super bowl or not
this is done sort of by whatever sort of happens out in the world one team beat the other team and that's why they ended up there but it's not controlled by the researchers it's not researchers that have flipped the coin and said
today san francisco gets there and new york doesn't
so it's still an interesting study and we'll get back to this example later um by the way and at the end of uh lesson three but it's not a field um experiment because it's not an experiment it happens in the field but it's not a randomized controls right
Constant Exposure to Field Experiments (NYT headline ex)
so a point i also want to emphasize is that we constantly expose the field experiments even
that we don't know it i told you how

amazon and google and ethics etc that run thousands of experiments every year and and you're also part of those you whenever you sign up to you you sort of accept the terms and conditions of these companies and of course you read those carefully right um you are accepting to be part of experiments and here where it's being written about the new york times they often in the morning will test different headlines so readers of new york times will be ex randomly selected so the readers some readers from new york times will be randomly allocated to different groups and for this particular studies the question the headline might have been which businesses should reopen first your favorite restaurant might be a super spreader is it safer to visit a coffee shop or the gym so basically the ones in charge of writing headlines they didn't really know which of these three might be the best and generate the most attention you can yourself think about one do you know which one would be the winner it turned out to be number three but this is a good example of a newspaper that realizes we can try to guess which headline is the best but why not run it expose let's say a thousand people early on in the morning for this 333 in one et cetera and 333 in the second and the third group let's see who clicks on the link the most and if a certain headline functions the best well by all means why don't we just use that headline for all the articles the rest of the day

this wasn't just the one
time thing this is something the new
york times does continuously
and i'm sure if danish newspapers are
clever they should do something similar
i don't know if they do
but um it's fairly easy in these
online settings to do this that's a
clear advantage of the online settings
so

Pros and Cons of Field Experiments
so just to summarize some of the
perspectives here we've tried to compare
lab versus field experiment i've tried
to illustrate
why and how we can run these field
experiments but also then
the pros and cons of this so it's clear
that it's more real behavior it's not as
artificial setting if you're in a field
experiment
the ecological validity is higher but
the examples then
of the sales insurance company also
illustrate you might have less control
over variables
you can't be 100% sure that the for
instance participants in the experiments
aren't
communicating with each other or that
something else might be going on
um so the randomization the clean
randomization is might
could be a little bit more difficult
because you don't know if people from
different groups are communicating with
each other
but it could certainly be a more
relevant sample it's not just
students that
study sales but these are actual people
that have to do the sales as
an insurance company and and you really
want to improve their performance
so that you have a certainly a relevant
you can have a more relevant sample
but it's also much more resource
demanding as you can see for instance

the the
insurance company study in total took
what is it
15 months or something at least the the
india field experiment on management
whether that matters that in the end
took at least a couple of years for the
first round then another decade for the
second round
these might be extreme examples but but
still
you have to convince the company of
running this and you have to set it up
and it's just much much more
resource and time consuming than doing
it in the
we have lab who who sort of are
going about their their life their work
life so they embed it in real life
meaning the stakes might also be higher
for the indian textile companies it
wasn't just
participating in a lab experiment and
then hopefully being the best
this was about survival this was about
making a profit this was about
which of the textile companies might
sort of be profitable in the long term
um so this is just sort of this is not
an exhaustive list this is just to
illustrate
that there are trade-offs a field
experiment is not necessarily better
it depends on the research question it
depends on what we prioritize
internal or external volatility
here i've just put in a question from a
former re-exam set in 2018
um where one was explicitly asked um
to compare two studies of that sort of
addressed a similar question one was the
lab study one was a field
study so so just to illustrate that
this has emerged in how this could
emerge in an exam setting
Survey Experiments
finally i just want to say that we also
do have other kinds of experiments that

are not sort of
clearly a lab or a field experiment we
could for instance do a survey
experiment which are
quite easy to pull off i know many of
you in your bachelor project on your
master thesis will do
a survey study but what is being done
here is you can imagine
that you want to ask about in this case
is um
something about fairness and earnings
and the original question was a 50 year
old man with vocational training is
working as a programmer and earns 1200
euros a month is this fair or not
well we could make it a 50 year old
woman or a 40 year old man
with or without education or we could
change the the
the occupation we can change the
question in many different ways and then
see
well let's say we do it with a 48 year
old man
is that would that constitute a
substantial difference in terms of how
respondents
will assess whether getting 1200 euros a
month is fair or not
or more anymore obviously we can change
it to a
woman or someone with another education
so we can
we can investigate a question not by
asking everyone the same survey question
in order to find out what people think
about a particular scenario but we can
ask them different kinds of scenarios in
order to find out
what really matters so we we would be
interested in vaccines and how willing
people are to take a vaccine
and then we can ask the questions in
different ways in order to really get to
a
better and more clear answer than if we
just sort of ask it in one way

um so this is also this would also be something that is possible to pull off in a bachelor project in a master thesis um if one um if if this fits the research question i for instance supervise a project where students wanted to sort of investigate uh the role of influences in social media and then they compared so what if an influenza that is either very well known or not very well known promotes a product that is very well known not very well known this sort of simplified version of the project they had and then they sent out a survey experiment where they manipulated some a few of the features of the survey and got different kinds of responses and then could provide an informed answer on the role of influencers in terms of being able to promote products we have to keep in mind here with these early experiments that they capture perception they can they capture what we think about this situation about the whether this is fair or not this does not mean that this is how we would actually act maybe we will have different reactions if we actually meet someone who fits this categorization same with the social media study maybe people will respond differently differently if they are on an actual social media site rather than answering surveys in the artificial survey experiment setup it's something we'll return to in in a later session but just sort of to emphasize survey experiment by nature what you get is how people sort of take a box which is very different from

a field experiment
where you actually want to see how
behavior changes
but this is just to highlight that yes i
make a main contrast between lab and
field experiments but we could actually
imagine
other kinds of other kinds of
experimental setups

<https://youtu.be/U9GNoIQ0qOw>

hi in this video i'll be presenting
different kinds
of research designs so in a second i'll
explain what a research design actually
is
and then go through three of these four
types of research designs we've already
covered experiments so i won't repeat
that
but i'll be explaining what's sort of
the key characteristics of each of these
research designs
and what are the strengths and the
weaknesses associated with each other
keep in mind here that the learning goal
is not just to be able to say
look this is how we define a
cross-sectional design or a case study
but to to give us the tools to assess
actual research being done that we read
about in a newspaper or a research
article or a textbook
and to assess the research critically
and to consider whether this particular
research design can actually answer that
question that was asked
this is what you also saw in lesson one
the example from 2018 exam case
the question five with brenda and we'll
be looking at that
example again in the study cafe in the
next lesson
this sort of this example quite nicely
illustrates the kind of learning that i
want us to
draw from this the research design

Research Design vs Research Methods
concept can be contrasted with
research methods so research methods
would for instance be
to engage in qualitative interviews and
and rely on simply structured interviews
to collect data
and then you also make some decisions on
how to analyze and code data afterwards
for instance
or it could be decision to uh
engage in survey-based research and and
rely on regression analysis to
to to to make sense of the data these
are sort of more
specific decisions of the particular
elements of the research that you engage
in
research design is an overall
frame so it's the overall decision
for instance to collect data twice both
in the beginning of your
of your project and at the end of your
project and in order to compare whatever
results you get
or it can be the decision to engage in a
randomized controlled trial and allocate
um a treatment to one group of people
and not do that to another group of
people
or it's the decision to look at multiple
cases in order to make sense of your
research question rather than just look
at one case
so these are some overall design issues
that one might engage in
so it's not whether to include this
variable or this case or
or sort of similar challenge it's
again it's on this more overall level
and
a study's research design can then tell
us whether the evidence that one has
collected
is then actually able to help us answer
the question that is being posed as
unambiguously as possible
so we'll see in the following examples

and the rest of the session
illustrations of this and how a certain
kind of study for instance a
cross-sectional study
might not always be able to answer the
question that the researcher has posed
so this is the research design angle
that we are looking at here
and and note also as stated here at the
bottom
the choice and quality of research
design very much depends on the research
question

again i'll get back to that

Overview of Research Designs

and again as mentioned it's the three
these three kinds of research designs
that we'll be getting into
and the learning again is to be able to
say

not just what research design is in play
in a given situation so

it's not just about identifying a
research design but also being able to
assess the strengths and weaknesses
of a given research design and for
instance if

in a particular case actually warrants
if the conclusion is warranted based on
the research design as
in play okay so let's get to it

Cross-sectional

cross-sectional research design the
analogy is all from this

snapshot idea you get um

you get insights just like when you take
a photograph you get a snapshot of
what's going on right here right now
so you would have more than one data
point you would for instance have data
on 100 people

at a single point in time and you will
know at a single point in time what
their height is and what their weight is
or or you would have insights on how
motivated they are and what dedication
is and what their age is and you want to
find out well

what explained productivity it's really
the core of many quantitative methods
courses
so the aim is to explain variation the
aim is to explain that on average
which variables for instance this
motivation education age example
which of these variables are associated
with the outcome
with productivity and we might find out
it's in particular the indication that
one has and motivation that that
is statistically correlated with
the outcome with productivity and what
this then can do
is to establish correlation but as we
heard
in a formal video correlation does not
necessarily imply
causation we might not necessarily know
whether it's motivation that causes
productivity
or whether it's because one is very
productive that one is also very
motivated to engage in a given task it's
difficult to
disentangle what comes first because we
only have data from one point in time
to sort of elaborate a bit of this
example and to highlight what we're
going for
so we would have for instance some kind
of spreadsheet with data
with data on person a and person b and
person c etc
we might know their education and and
motivation variables et cetera and their
salary or productivity
and then we can do some kind of
regression analysis to find out and
on average how does different types of
education for instance
impact salary or productivity whatever
we're going for
we're not really interested in each
individual person
as such we're not really interested in
why a particular person

chose a particular education ended up
with a job that
landed him or her a particular salary
we're interested in the average
variation
that the associations between different
variables
lead to so it's the average course that
we go and go we're not going for
understanding a particular person
but but again when i say cause we should
keep in mind it's it's always
cross-sectional it's it's it's
it's difficult it can often be difficult
to establish that
um it's motivation that leads to
productivity and not the fact as i
mentioned before
the fact that one is very productive
which then leads one to
become very motivated as well
Longitudinal
so this then stands in stark contrast to
what we call the longitudinal design
so in cross-sectional design we would
for instance compare 20 and 40 and 60
year old and we'll
carry our research in the year 2000 and
we'll see
whether each of these groups are more
motivated or more productive or whatever
our issue is
in a longitudinal setup we would study
in this example here the same 20 year
olds in 2000 and then when they were 40
2020 and
when there are 16 2040 we would study
the same participant
and vary at various points in time so
this is really the key thing here is we
have
data over time on the same people we
don't just have data on
some 20 year olds and then in principle
we could collect data on other 40 years
old
a couple of weeks later this is not what
we mean we have data on the same units

at least twice as highlighted here so we could for instance have data on the motivation of employees the time point one let's say in january 2021 and then we could look at how motivated and productive they are in february 2021. this might enable us to say well it's actually a motivation in a prior time point that explains productivity at a later point in time so this improves our chances of establishing some kind of causal connection um because we're going to identify what came first in a cross-sectional setup we can so because it's the snapshot it's a picture taken at one point in time we can't tell what came first it's just it's basically not possible at least not if we just go by the data that we have um in a longitudinal setup we would be able to say well since we have data from two points in time we can say that something comes before something else otherwise longitudinal research can in many ways be the same as cross-sectional it's just adding an extra time point of course it can be you can collect data from the same people every week for a year or every month for a year or every hour for a day as long as you have multiple um as long as you've collected data from the same whether it's individuals or firms or cases or countries or whatever it is at multiple time points which because this enables you to compare development over time and it will potentially enable you to say what happened in time point one is actually explaining what happened in

time point two
we'll get a few more examples in that in
in
in this lesson four and this can be done
qualitatively it can be done
quantitatively by sending out the same
survey multiple times
but it can also be an individual who
actually observes
a group of people over a long period of
time and this can also constitute
longitudinal research
then as a as a third one presented here
Case Study
in this video we then have what we call
case study design
which is of quite fundamentally
different nature here we are looking at
some kind of
bounded situation it could be a unit in
an organization it could be an
event um it could be a person
that we are following or multiple
persons that we are following
and here we're really interested in the
in-depth reason for whatever is
happening so before i said explicitly
in the cross-sectional study we wouldn't
really care about why peter did
something we just look at
how various variables for instance the
motivation and age and education of a
bunch of people relate to
their productivity and we don't really
care about the individual as such
only in the sense that we want to get to
an average result for instance vr
regression analysis
while in the case study design we're
really interested in the individual
persons and why is it that peter
um has the particular education
and has this why is it that he's
motivated in the way he is and why does
it lead to a certain kind of
productivity so we don't really care
about the averages of a lot of people
because because we don't have

data on a lot of people um
we we want to focus on the con this
particular context the
unique context we're looking at this
case study design doesn't have just to
look in one case we can look at multiple
cases
um and compare them again we wouldn't
sort of do any kind of statistical
analysis usually
we would in a qualitative sense trying
to compare
differences between different cases
maybe we have a unit in one country and
a unit in another country and want to
find out what the differences are
between these units
case study design can also be
longitudinal we can also follow a case
over time and thus have
multiple time points multiple data
points on the same people over time
but again not with the aim of
understanding some kind of average
correlations but understanding why stuff
develops over time so it's really how
and why something is happening that
we're going for
typically this could be interviews
observations but you can also use
surveys or archival data or other kinds
of approaches
so whether something is cross-sectional
design or case study well depends on
what you're going for
um you might be comparing 6 8 10
12 different kinds of cases but what
what answer are you looking for are you
looking to try to
understand
each particular case and what is
particular about each case
and then what happened in each
situation or are you going for some
average
variations and and sort of caring
less about the particular context
so as mentioned here then we can also

have single versus multiple cases
and um and there are some trade-offs and
advantages of each of them if you're
doing either single multiple case study
you can read this paper that provides
more insight into
the pros and cons of each approach the
more cases you're looking at
the less in-depth you can look at a
particular case i mean basically think
of it as a
resource question you have the resources
to carry out let's say 10 interviews
well you can do you can take 10 cases
10 individuals and interview them all
once you can look at
five cases and interview them twice over
a month or two
or you can take one person and interview
this person once 10 times or for
instance once every week over a 10 week
period
so it's all about how you allocate your
resources and and the more you focus the
fewer cases you focus on of course the
more in depth you can go with this
particular case

Differences

so they're really trying very different
things
um the cross-sectional study is doing
this
snapshot while the longitudinal study is
like the video camera that is
um is tracking how stuff develops over
time
while a case study can be longitudinal
you can follow a case
over a long period of time but you don't
necessarily have to i mean you can do a
case study where you just
go out one day look at what happens in a
company
and unless you're explicitly aiming for
the difference between for instance 10
o'clock in the morning
and three o'clock in the afternoon we
would usually call that one

time point or something you're going out there to collect data once and trying to understand that so it becomes longitudinal the moment there is sort of something you're interested over time that you're interested in in differences over time something happened maybe new employees were hired in the company that you're you're observing or doing interviews at and you're interested in how that changes their responses if we can think of it in this qualitative case study setup

FAQ and a Short Test

i have here in the following slides provided some answers to some comments or some typical questions i'm getting throughout the years i won't sort of go through it and read it it's it's you can read the bullet points yourself and it's just the answers to some typical questions that i would usually get and it also relates to some of the points that i've already talked about and i've also inserted very sort of simple multiple choice uh setup where you can look at these questions and the answers are the next slide just to see if you're really getting the the basic elements of this of these different designs right

Differences (continued)

so just to jump back to this slide here in order to assess whether something is for instance cross-sectional longitudinal try to pay attention to how is data collected is data on the same person's being collected multiple points in time so it's not enough that you have some 20 year old and some 40 years old you actually need

the same people need to collect data on the same people twice
um so also a survey that sort of goes on for a week where you get one response from from each respondent just because the survey sort of takes i mean it can be responded to over a period of of a week as long as people have only responded once and and this is what you're aiming for well then it is a cross-sectional design and in a survey you would usually then only be looking at sort of average interactions between different variables as you will ask people well on a scale from one to seven what do you think of this or that while in a case study you won't just want to get these very generic answers and generic sort of ratings for instance on a leakage scale but you'll want to ask the person well what is it that you're thinking about this and then why is it that you made this choice and what makes you rate this question in this way so you really try to dive into the details of people thinking we'll get more good insights into sort of further challenges to to various kinds of research designs but also get a bunch of examples that makes a lot enables us to get a better sense of what these different designs can do and what their strengths and weaknesses are we'll be getting that to in the live session as well

<https://youtu.be/iVBvrr48kDQ>

in this video i will be addressing some further research design challenges that research might be

exposed to so while the research design
2001 text is
this classical textbook chapter it
provides an overview
and some general strengths and
weaknesses then this amg 2011 text
is on a more advanced level and trying
to provide
insight into some additional and
challenges that research design might
might face
so in an exam situation don't expect all
of these to come into play
but you might have an opportunity to
identify that one or some of these
challenges are in play in the
description of the research design
so again this text is written for
researchers
um and it's trying to outline what are
some typical pitfalls and typical
reasons for
why a submission of a paper to ace one
of these world's leading scientific
journals might be rejected
so it's trying to outline some some
typical
um flaws some typical things to pay
attention to and i'll go through all
each of these in in the coming slides
but also note what they write here that
there are no
hard and fast rules to apply here so
matching research design to research
question
is as much as art as a science so these
rules are not completely clear cut
sometimes they can be ambiguity but it's
important to know about the
um these criteria and these potential
pitfalls
because it can help us develop stronger
research designs and it can help us
identify
a given study that we read about whether
it's a newspaper or a textbook or a
scientific article
it might help us say look they actually

overlook this or due to this particular reason we should have less trust in the results

so let me try to go through each of these elements here

Cross-sectional Data and Change

so one of the main elements that i've also

addressed in the in the form of video is

that if you have cross-sectional data

it is difficult to address change so

let's say

that you have data from 2015

and you have data from thousand

companies and you know which companies had an

increase in productivity and you know

which companies had an increase in

salary

you will not based on such a

cross-sectional data set where you only

have data from one point in time

it is difficult to know whether it's

productivity that led to an increase in

salary

or whether the company offered an

increase in salary

and the expectation of this

then led to a later increase in

productivity it's difficult to

pinpoint the direction of this arrow

here

when it comes to sunshine and eating ice

cream we know that eating more ice cream

doesn't lead to more sunshine

so we know it's this way around but

often in in organizational context and

business

context we might not really know in

advance

what the direction of the error what the

direction of the causality is

in this case if you had insight into the

productivity in 15 16 and 17

and simultaneously if you also had

insight into salaries

you might be able to see that salary

increased

first salaries increased already in 15
and 16 and then only in
at the end of 16 and then beginning of
1718 productivity began to increase
let's say you had data on
on a monthly basis in this way you would
have longitudinal data you would have
data on multiple time points
and you would have a stronger foundation
to
be certain that it's actually that the
arrow has a certain direction of course
you could also do an experimental design
have a thousand companies 500 of them
will increase salary 500 will not
and then see if that influences
productivity
there's also some fancy statistical
analysis you can do this is not
something we'll be
getting into but this is just one of the
general points and that we often see and
also this if you remember from the
causality correlation video
i talked about this overview of 50 most
shared posts on social media
this is what often happened is that they
will have data on one point in time and
then
end up making a causal argument even
though that there could be many other
reasons for
why we see a correlation
Omitted Variable Bias
next element to look at is what we
call omitted variable bias
this is when we have the situation where
important variable
where important data is not collected
and this could then
lead us into a wrong direction so
um at the bottom here i've outlined the
student bookshelf and future salary sort
of maybe a little bit of a silly example
but it still
gives you an insight we might be in a
situation where we can say
that the more books there are in

a student's home the better the academic achievement
the student will reach this is a correlation that is very likely true the more books there are in their homes over of a student the more books there were in their home when this student was was growing up
in at his parents house the the better and the higher the academic achievement and but but this kind of sort of setup might make you think well let's just put more books into the house of um of families because then more people will end up have a stronger academic achievement
and that's probably not the right way to think of the causality because it's probably the parents intelligence or educational attainment that decides whether there are many books in the home
and it's also the parents educational level and the sort of the parent push towards the students future education that then influences academic achievement so yes parents influence the number of books at in in the home of families but they also and probably more importantly in other way influences the academic achievement of students so this is a example of if you'd only collected data on number of books on academic achievement
you won't get to the actual variables that really are important
one could also imagine a an example of kovit 19 and we could look at different countries and then see how many people die of kovit 19 and then we could sort of um that into a conclusion of well certain countries are better at preventing kovind 19 death than others if we don't

consider that different countries have different demographics so different countries have more less number of old people and the fact that all the people are more at risk of dying of covid 19. well then we risk making the wrong conclusion that certain countries are better at preventing covid 19 while the main reason actually could just be that some countries or some areas simply have fewer people of old age which is then the reason why they have less people dying of covid 19. so i could sort of continue to make lots of more examples the point is that we should be cautious about trusting any relation between x and y until we've made sure that there isn't any other variable out there that is correlated to this and it is actually playing the key role in the example of the elementary school study and how different points of time they had an influence on how well they did the test well if it turns out that the students that were really really good also got the test early in the day well then you've omitted an important variable which then influences and skews you and destroys your conclusion so again it's always important to try to have data on all the relevant variables and this is then also again the advantage of an experimental setup because you can control that there isn't any sort of random or systematic variation sorry there isn't any systematic variation going on that we can't control so omitted variable bias we often see inappropriate samples also see inappropriate samples being used so let's say you're interested in

how
top management engages in decision
making and then
one might run a study for instance a
laboratory experimental study where you
used regular students
to try to dive into this issue well
that might not help understand how top
management makes decisions because these
students are not by definition yet
top management so maybe you could use
mba students but actually you really
want
data from csos to really make sure that
whatever you
end up finding out really matters for
how top management makes decisions
um
one could also imagine a study on
whether face masks actually
helped prevent kov19 and if this study
has been done on
on nurses and doctors and then shows
how well they
are able to use mass to prevent getting
coffee 19 well that might not
necessarily translate to the rest of the
population
who doesn't have the same training in
using masks um
so again one would have to make sure
that you don't just sort of
transfer insights from how specialists
are able to use mass to the everyday
population
so make sure that when one assesses
the conclusions of a research study or
looks at the data
and ensures that whoever is sort of the
foundation in the data set what
the data that you've collected and the
people you've collected data on that
they're actually also relevant for the
conclusion that you're trying to draw
another example of this is a slightly
different example
is is google and google the company has
realized that

general test scores what they call sat scores in the united states or grade point averages don't really predict how well google employees perform at the job so it's just not really particularly helpful when google looks at all the let's say a thousand people that hire every year um the grade point average of these people from their college degree does not really help explain who performs well at google and then one could think well should a typical danish firm then also just ignore grade point averages well google only hires employees with very very high grade point averages basically if you are from a danish institution you more or less need an average of almost 12 in order to get a good chance of getting a job at google so there isn't really any variation in jp grade point average to take departure so of course if there's no variation in the variable that we're interested in this variable is not going to help you explain how well people perform similarly if we look at the height of nba basketball players in the united states they don't really predict scoring very well this does not mean that height doesn't matter of course height matters it just means that almost everyone that gets into the nba are already very tall which means that this factor is not going to play a big role so um typical danish firms should probably not hire like google because the variation in the kind of people that you get in and the grade point averages it's much much bigger so one should not necessarily look at google and amazon and

apple etc to see how they hire people
and just copy their
their hiring approaches because they
might not really translate very well to
other kind of settings where you get
other kinds of
applicants this
Selecting on Dependent Variable
is also related to this issue of
selecting on a dependent variable that i
already mentioned once
so let's say we're interested in
how successful entrepreneurs they do
well
and we end up only studying successful
entrepreneurs
well we might not really find out that
unsuccessful entrepreneurs might do the
exact same thing
maybe they try to engage in most of the
same practices but just fail for various
reasons whether it's luck or
or a factor that we have that we have
omitted so
if we want to explain how a certain
group does well
entrepreneurs or basketball players or
ceos or
businesses or whatever we're thinking of
we should not only look at those and
actually do well
because we risk missing some really key
information for instance that those that
don't do well
just end up doing the same thing in
principle there could be industries
where it's pure luck that decides
whether you succeed or not
and um and you'll only find out about
that if you look at
both those that do well and those that
don't do well so always pay attention to
in a study
if they say you should do this or you
should do that well
who have they looked at in order to
reach that particular conclusion
you can also imagine a study that says

look at all these companies that
successfully manage the kobe 19 crisis
this is how they survived
well again look at those that might have
tried the same thing and
failed and so don't just select on the
dependent variable don't just select on
the outcome that you're actually
interested in
but make sure that you have variation in
that outcome

Careful Construct Creation

then the article also talks about the
construct creation or what's the sort of
the relation between the data you
collected and the actual theory
so and again maybe insightful to refer
to the face mask example that i've given
before
if you claim to have measured the
effectiveness of face mask
well it might not be ideal it's not
really ideal
only to actually get data on
whether people received face mask in the
mail
and an encouragement to use them that's
not directly measuring the effectiveness
of face mask
that's measuring the effectiveness of
encouraging people to use face mask
so the theory should not be on the
effectiveness of face masks the theory
should be
on the effectiveness of encouraging
people to use face mask
and and the same thing with all the
range of other elements that i'm
mentioning here always pay attention to
if you claim to have measured employee
performance well
what is actually being measured here is
the manager's subjective
assessment of it is it some more
objective criteria that is being
followed
and saying with firm performance it can
often be measured in many different ways

so one wants to be pay really attention to that the concepts you use in your theory and the words that you use in your theoretical framework also match the actual data that you collected

Common Method Variance/Bias and then there is one mention that is probably a little bit more um complicated to get your head around something called common method variance common method bias it has both of these names and the point is that you should be

careful for instance in a survey setup if your independent independent variable stem from the same source let's provide an example let's say you want to investigate what employee characteristics leads to high job performance

we could have a setup where we send a survey to managers that are to rate both the characteristics of the employees and the performance of the employees

but this then could create this common method bias because it could be that the manager's particular thoughts on useful characteristics maybe a manager in particular likes a certain gender or certain education or a certain attitude and this

then also leads the same manager to assess the performance of these employees in a particular world so it's actually just the manager's liking of a certain kind of quality that also leads them to rate the performance higher even though the performance in

actuality is not necessarily higher so we have the risk of a common method bias because the data that we're getting on both the independent and the dependent variable

is from the same source
we risk that it's systematically skewed
by
the same source assessment of both of
these elements
so if the source is biased the biases
influences both the independent and
the dependent variable in the same
systematic way so you might end up with
something that is very
significant and looks very substantial
but it's just an artifact
of the way that data is collected so
just then briefly put it's much better
to have an
independent source on each of these so
ideally you would have
one for instance one manager or some
range of people assessing the
characteristics of individuals
and then another range of people or
other managers
assessing the performance of these
employees because then you don't have
this
systematic overlap in where your
information is coming from so
ideally you have multiple sources to get
your data from
it just leads to it stronger
just leads to a stronger data set a
stronger design
and a stronger ability to get to the
conclusion that you actually want to get
to

Summary of Research Designs

so these two videos have tried to give
an overview of overall research designs
and strengths and weaknesses of them and
also some
particular more sophisticated elements
that the amj article goes into
this amg article also goes into topics
that you will be
required to get a have an understanding
of
to already have an understanding of when
you are at the master's level so that's

why i'm also already pointing to them
here and here the last line i've just
given some very general and some very
rough rules of
thumbs of advantages and disadvantages
of the various kinds of research designs
these are really
just very rough it's not always the case
that a cross-sectional
design will have a strong external
validity and etc etc it's not always the
case an experiment will have a weak
external
validity but these are some sort of
general rough characteristics
that can help one make sense of
different research designs
in the next session and in the live
session they will then be focused on
quite a few different examples of
different research designs and how they
can complement each other
in order to get a sense of what do they
do what are they and and how can we
assess the quality of these so that will
be
the focus in the in the next session

https://youtu.be/kH_BXsAmEt8

they present an exercise and some
remarks on two of the exam questions in
the 2009 read exam the question set
first an exercise inspired by current
events so again as before as in the
study cafe I suggests that you'll read
the information here there is some
information on what happened in two
different regions in Italy and down here
on the x-axis
yes time and on the y-axis we have
number of Corona cases and the graph
indicates what happens in these two
different cities so I suggest you spent
a couple of minutes thinking about
randomized control trial cause I
experiment something else what's going
on here so after you've had the

opportunity to think about this for a little while having posted I here present some thoughts on it so what do we see we see that two different cities areas in Italy made different decisions in terms of how to handle the coronavirus lorry engaged in social distancing pagamo did not and we see a clear difference in terms of how the number of cases has developed over time is it a randomized control trial well the data does show clear difference between the cities it's argued or the way it's portrayed is that there's sort of one variable that is different we don't know if these two cities took other choices we don't know if they did anything else maybe it's not only social distancing so so yes the cities are different but maybe also the people the cities are different maybe better if a government just has more old people maybe low he doesn't have maybe it just has far less there could be different reasons in place so we can't really call it a quasi experiment because cities have self-select into these different treatments but it's still the data set that strongly indicates that there could be something going on about this social distancing the value of that so it's certainly a longitudinal data set that shapes development over time and it's potentially shaped by this one key difference calling it a randomized control trial or quasi experiment we can't really because we don't know how many other things have been in play here okay next up is then question 1 from the 2009 we exam and again I will suggest that you spend some time on trying to read this case read the questions as outlined up here there are two different studies and try to think about how you would answer these questions and then in a minute I'll I'll provide some thoughts or non of this so if we when you've an opportunity to call it think about it we

can move on you to the next slide where if for one I've just highlighted some key elements to pay attention to I have some more remarks on the next page but just want to highlight a couple of the key things so here in study one its clip made clear that the participant decides when to show up based on their own assessment of their sleep and some students show up having slept well the first time others lip roll it the first time they showed up so there is not a randomized allocation of who sleeps a lot and it doesn't sleep a lot there is self selection into when you show up so the sleep number of hours you've slept hasn't been manipulated and so some one thing to worry about is well what might be the reason for why students live boy maybe they were stressed and worried and have stuff to to think about which might cause and sleep poorly which might then again cause them to perform poorly on whatever tasks they were supposed to to carry out so we don't know if it's an effect of the sleep because we hadn't manipulated it as we would like to do in a randomized control trial so again there is this self selection into the two different treatment or whatever we should sort of sort of call them so there's an on average variation there is focus on comparing across two different kinds of groups there was no long eternal day such we're not comparing before and after we're just comparing two different sets of students who have slipped either a lot or a little we we don't not a measurement of what happened before something after something so it's cross-sectional design and let's sort of the overall assessment of a word that we want to call it an RCC we could make some further remarks on the internal validity and I've already mentioned this that we don't know why they sleep poorly it might not be an effect of sleep it

might be an effect of something else but there's also something about the measurement of sleep where we can see in contrast to study to the measurement of how well people have slept is not as strong do we always know if you slept poorly or well I guess we have an intuition and some might have smart watches that captures it and then we haven't sort of taken this into account maybe some of the people have a very good measurement and and maybe others won't

so on top of not really quite being an RTC on top of there might be an element of people sleeping poorly because of something else meaning that there's something else potentially the stress or the worry or whatever it is that is actually causing variation on some of that we also have a poor measurement of the contract of sleep that we interested in and again on the next slide you'll see more I'll just move on here just to sort of let you know there is something more coming here about some of the remarks I'm making here in terms of external validity it's it's sort of unusual in compared to the examples we've seen so far that they've actually done this study and in four different labs around home at around the world but at least in Europe and North America so that's stronger ability to generalize from the sample they've relied on to a broader context it's not just so one particular narrow geographical area but they were still this interaction of of setting and treatment so it's a particular kind of task and dealing with conflicts in the lab is certainly quite different than dealing with conflicts in the workplace in an actual organization so there is this interaction of setting a treatment and and there is also interaction selection and treatment of the kind of people that selected into this in part because of the areas that

come from but also who actually does take part in studies at universities they might also be particular so we have this couple of reasons for why the external validity should be sort of limited and then since it was happening in a lab and as usual there's also a weak ecological for the disease you don't have a lot of information about the task they had to do so difficult for us to say a lot okay so study two um here we have an allocation of half the participant into one group and they sleep less and the others sleep more don't have explicit information about a random allocation but we guess we could assume that just point it out but but at least there is a clear treatment and certainly stronger internal validity then it's 31

I guess one could say why were they woken up during the night why couldn't they just sleep less and then we woken up after six and a half hours so there are different ways of implementing the treatment sleeping lists there's actually variance just like Nelson would say there are different ways of measuring if one I slept less when you wake them up every three hours to just make them sleep less in the end just another example of this heterogeneity and external validity we have the same threats as above and we don't be only even though you had one lab so you don't have that variance across the lab and sleeping in a lab is certainly also very unusual there might be some kind of automatic that's also in play here but your overall surgery stronger internal validity because you've had this allocation of people into different treatments it's mentioned we have overview of some of the comments I've made here that you can also peruse and get a handle of what are the key elements to pay attention to okay another another question so please read

this think about how you would answer this and I'll provide some remarks afterwards which are also highlight slide okay so we have this quote from the physicists hiding back and as you will hopefully notice this is the quote that was also shown in class not just this year but was actually also shown in class last year it wasn't a surprise for them to to see it and then the disa didn't use some of the remarks that you could offer here and please rely on the curriculum to expand upon and explain what the statement means and we could for instance say there was something about this we don't observe nature itself but nature exposed so apparently there is something that can be exposed there is something out there to be revealed it's not we don't reveal nature by itself but we could start whatever sort of way you could finish the sentence so something to be exposed so sounds very much like those positivism and scuba says although a real world riven by real natural causes exists it's impossible for humans truly to perceive it with their imperfect sensory and elected mechanisms does sound very much like the groats right so it's to me logically speaking as Gruber says it's some sort of modified objectivity we cannot obtain truth object with objectivity in some kind of absolute sense and then an answer one would then expect you to expand upon this post positivist position and why it's not the other positions why isn't that positivism or constructivism fits but but this is sort of the key elements to get to and some of the key perspective from in terms of the isms to invoke we could also rely on come and say well this phrase the method of questioning corners talked about how we have certain paradigmatic ways of trying to understand the world within different paradigms we have beliefs values

techniques etc we have these four different elements of the paradigm and and so we don't observe nature as it is in itself because we have paradigmatic glasses also to speak we we have some expectations and education and experience that shapes whatever we see out there interpret so one word one wants to expand this a bit more good answer what we would talk a little bit more about what a paradigm is and how and for instance the throat about how we literally don't see the world as it is in itself but good interpretation always ends where we're seeing interpretation begins when when seeing ends so you would you could also emphasize that quote finally we could also draw upon Nelson and how he says there are some genius variables that fuzzy boundaries don't have the RIC access to nature as it is we have different ways of trying to measure it whether it's makes you understood as unemployment or motivation or research and development in your firm performance so sleep or whatever it is so the number on the assumptions behind its construction for instance the number of how we measure performance whatever it is comes to between us what the phenomenon is rather than as a useful if limited measurement of that phenomenon so it's sort of the similar point of view about this expose to first time that we don't get the a sense of how the world actually is in itself even though that's Mike what we think we treat it as if we see the world as it is in itself but as he says it's a limited measure there was some these are constructed measure that

<https://youtu.be/62QVoC7QBG4>

released a couple of questions on experiments what is a natural experiment what is a quasi experiments what is an

RT C's that I decided do a quick video on this I would like to add though that it's not a key aim of this course to be able to tell for instance the difference between a natural and a koala night experiment the key aim is to be able to say look this these are the characteristics of a true randomized control trial this is proper randomization we have identified a causal effect versus a situation where and then a setup doesn't quite live up to these requirements and thus turns out to be quasi quasi experiment not a quite a true experiment that's the key goal just to reiterate what is it that an RTC is first of all we are really keen on this randomization we need a randomized allocation to each treatment we start out with a bunch of people whether they're all male Howard students as in Donna Riley study or Chinese factory floor workers or whatever they are we just want to try to make sure that there are as similar as possible after the randomization we know they're not going to be completely identical people companies are different but we want them to be as identical as possible we could even do some some post tests where we we for instance make sure there was an equal number of men and women age group that have the same age etc okay so we want this randomization to have worked out and we want two groups to be as identical as possible and we want this to to continue to be the case and and we also want them to be in as control and similar environment as possible we we know that if we put people into different classrooms and vengeance change the teaching styles that while the classrooms might be slightly different but so we're going for this this aim of having no systematic differences between two different settings and we want these very controlled environments because we might

want to make sure that the two different groups are similar as possible in a similar environments as possible and only exposed to one particular treatment whether it's how they built Bionicles of whether it's the curtain or whether it's a time of the day where they engage in tests any kind of treatment we could imagine to make sure that there is only this one difference between the groups that we have in mind and in a quasi experiment what we're then trying to pay attention to is to say well okay we realize in the in the elementary school example about cognitive fatigue for instance we realized the location wasn't randomized as such it was people employed at various schools around the country that tried to make sure that that there was classrooms available IT equipment available and that's why sometimes it was for a that had the the early time and sometimes it was the class for PE at the late time but we could then argue that the process was as good as random because kids didn't self select into I want to do the early one today or school teachers didn't self select their classes into certain times of the day it was just as good as random a process and and again in the course I experiment if it doesn't happen in the lab or we have complete control over everything we can't be hundred percent sure that there is only this one difference but but hopefully in the end we'll have been able to identify the one true one key factor so in to take the example of the Spanish flu example again we we had that the Spanish flu example when the x-axis have how quickly cities and each circle is a city how quickly cities responded to the crisis and close down schools on the y-axis we have the the impact of the flu hat on whatever Chi in in the given cities we see a fairly strong correlation here zero

points 37 that's far stronger effect
then the headache pill has on the
reduction of headache question is is
this a quasi experiment can we say that
there's only this one variable that
makes Pittsburgh and New York and st.
Louis in Chicago it's 0 different well
actually if we look at the original
study we can
say no there were some interventions
that the city could engage in and there
were school closure cancellations of
public gatherings isolation and
Quarantine so there were actually
different things in play here there
wasn't just one variable that
distinguish these different cities so in
an RTC we can actually claim look it's
always this one thing in the elementary
school example since we had a large data
set and since it wasn't the kids
themselves that made any kind of
decision we can be reasonably certain
that this is the key factor the time of
the day that took a test but in this
example what we can pay attention to is
that was clearly cities themselves that
could self select whether their own
school should be sort of into one
treatment or not and that also makes it
clear that there could be lots of other
things that we don't know about again we
could go to the original study and find
the true answer here but already in
principle we could be a bit more
uncertain about this so in any kind of
so naturally occurring experiment
whether it's an elementary school the
Spanish flu trying to pay attention to
who are these so what these people or
schools or cities or companies that are
in different treatment groups how did
they get there
and then a natural experiment it simply
then defined by it was something that
happened in nature and the elementary
school example is then not quite clear
on it wasn't research as it did it but

but yes so an existence would showcase something that clearly is something we would call a natural experiment it's not quite a true randomized control trial in the sense that we can't be hundred percent certain that there wasn't some kind of systematic factor not a hundred percent certain like when we flip coin in a lab but we do notice here in this example that we saw and study good thing one that there was variation in the amount of nuclear fallout caused by the Chernobyl disaster and there was this reach

variation and the regional variation depended on rainfall it turns out that depending on how much rain a given area got it got more nuclear fallout and this is an leads to variation in standard I'd cognitive test 25 years later I have a longitudinal data set seems to be a treatment and the treatment is how much nuclear fallout do you get as a person living in Germany at that time as I said I lived in Germany there and 86 so it's not a formal randomization process but I think it's a situation where we can be fairly certain that this probably is as good as random the link between cognitive ability and variation in that and and D and whatever happened sort of early on time we does seem like that we can say the Chernobyl disaster was a key factor the region's various areas in Germany they didn't decide himself whether they should get rain fallin on it was an external shock it was an external factor which is why we can call it a natural experiment we would still usually call it some kind of quasi experiment because again we can't be absolutely 100% certain that there isn't some kind of relation between whether you're in agricultural area or whether [Music]

or whether there could be any other systematic reason for why cognitive ability is linked to these different

regions and the way it is so again natural experiment it's something that sort of happened outside the researchers control of in nature in this situation with the rainfall we can consider it basically as good as a random but we still do prefer in the proper randomized control trial to have absolute control of the situation and be certain this group or this treatment that group didn't get the treatment and there's only one difference between these groups

<https://youtu.be/L15ZJ8OmpQA>

a brief note on research designs that also includes some answers to questions about Bernstein cases and randomized control trials so we have these different types of research designs that we can identify and we know their strengths and weaknesses associated with each of them cross sectional we could consider is like a picture it's it's a snapshot you have a picture of something right now this picture could entail asking kids about how many hours they played video games over the last three weeks so but it's still a snapshot of what they think right now it's not actually collecting multiple data points so just like a picture of cross-sectional and we have longitudinal with like a video camera or having multiple pictures taken you can see how stuff changes over time for instance then so you have multiple data points you've asked one months ago about you asked kids about how many hours week they played video games and now you're asking again today and you can now get the difference you're not just getting the perceived difference of what they think what they did 1 months 1 year 10 years ago you're getting insight into how their answers actually develop over time the experimental approach is then the the treatment someone gets the

treatment the match that is on fire and the other one is not on fire and that leads to different kinds of outcomes and you would then of course have many different ice cubes you would do this on in order to see is this not just a random fluke in some sense you could say that it experiment is always about overtime if you start out with two identical groups and then you do something just something different to these two groups and see how that sort of what happens then after that but you can even as in the Bernstein situation have an experiment that runs over five months and then see how the difference is develop over time

case study is then not just a picture it's it's it's actually understanding something about a big specific case a specific context this could be one individual company this could be a group of individuals group of people playing video games together for instance so a case is is so slightly different and more tricky to define but it just it focuses on the context it doesn't focus on the the variance within Howard thousand different people answer a survey it would focus on the particular people and their particular answers and why they gave these answers I should add that a case study can be an experiment as such we can have an experiment embedded in a case as in Bernstein there was a large Chinese factory and different units than either were behind a curtain or not but you can't run an experiment on one person on one sort of single firm then you would have to run it on people who lots of people within the firm has to be something you can compare to if you only have one individual you can't in terms of how for instance to address if there are multiple research designs other than actually two different studies we should assess differently and or separately and

and I would say no I mean you an overall research study can consist of different research designs and you can then first identify well I think it's a field experiment first foremost but has no matutinal elements that is actually also trying to understand a particular case but and then that's it the important point here is when you then assess the quality of the research design you shouldn't consider them in isolation yes there are some weaknesses related to the field experiments approach to answering whether something has an effect or not we don't really get insight into why it has an effect this is an the case based interview approach could do and you have to assess the overall research study that burns time carryout you have to combine these two things and you soli considered in isolation you can of course say the weaknesses of one thing one approach can then be alleviated can be sort of can be corrected so to speak by by the qualities of for instance here the case based interview approach so consider it sort of from an one overall one overall perspective of what's the strengths and the weakness of Bernstein's study identified that there are multiple research designs in play and then make that lead into an overall answer to the strengths and the weaknesses of the study that bench time carrier

<https://youtu.be/LQ6ZLZoG0o>

hi everyone I want to do a short video on causality and the challenge of identifying if there actually is a causal effect it's really been a key aim of this entire course that's why we've learned about experiments that's why we've learned about research designs in general and try to improve our ability to become

better at identifying what can a study actually do because it's the aim of much of Sciences to be able to say look this kind of management style actually improves performance implementing a curtain in the Chinese mobile factory actually changes something so this has really been a key aim that we want to get how can I study create a causal or claim a causal effect and what does it mean to say that something is true or something is a true story about something and and really a key challenge is also that in the media and journal we tend to see studies being overblown exaggerated and this is actually what what this slide is particularly about and showing a lot of stuff going on is let me try to explain it so we have the 50 studies that has been shared the most of all the studies published at this outlet they have generated 1.3 million social media shares so these are scientific studies that have been shared a lot on Facebook Instagram Twitter etc and what is then showing is that these the authors of this study they looked at the 50 studies and try to see how strong the cause of claim one could make so if you're all the way out here to the left well then it might be just any kind of correlation it might be small sample size there's really a substantial risk that there was not an opportunity to say that something causes something else and we can't say if the study claims that video games causes aggression or if it's just a small sample correlation and poor collection of data well it's just not a strong claim the further we go to the right the more likely that's what the authors of this article claimed more likely it is the causal inference is generally well conducted and reported might be some issues and that we can't be under percent sure but even over here in the very very high section that says causal inference is extremely likely to

approximate the results of the ideal randomized control effect so so so even out here it's saying it's not necessarily perfectly true and the causal effect but at least we can approximate this is as good as it gets and as you can see most of these studies that have been shared like 1.3 million times actually only very few of them out here on the right three are in these two categories most are far further on the left actually don't allow the strong claim of look something cause or something else okay this is the categorization of these 50 articles then we look at what team articles of these the authors of these 50 articles themselves claimed and it turns out a little bit more positive to Mystic about their own results and who knows who who is right and then that's not the main there are more optimistic they think thinking so they have stronger claim to causality but the key thing is really to notice a difference between what the media says that's the bottom line the media says that 44% of all articles can claim I have a really strong claimed causality it says something like is caused by is due to statements without room for doubts or in general but even the moderate ones here is that well may be caused by seems to result in causality is suggested which is sent in the end 39 plus 24% almost all studies are quite strongly hinting at causality are claiming it even though if we go back and look at the original 50 articles it just doesn't seem to warrant such a strong claim the data does not seem to warrant the strong claim that we see here so in a nutshell that's actually she challenged that I want her to know about this is one of the key things that I wanted us to focus on and all about that this does happen this is a problem we see overblown causal statements in the

media and I wanted us to become better at then understanding when something can be a causal effect and also which kind of research design that enables us better experiments field experiments longitudinal study carefully design correlation studies that might sort of at least have a better shot at not being absolutely on the left but can sort of push it to the right that's always what we want to do push it to the right on this overview because although and also indicate before we should not think of causality as a binary thing it's not just true or false it's some claims are stronger than others it's not true or false as such whether having a darker skin color leads to a higher risk of getting a red car but this explanation is certainly getting us there the internal validity is fairly strong but what we should also think of the explanation the causal explanation in terms of external validity it might hold true for these European football leagues you don't know if it also holds true for all kinds of other sports or in the future etc but and then also related to this one might go away from from what we've heard so far and say ok so apparently we need a randomized control trial to really be able to say that something causes something else and and and true it's a golden standard it's something that really enables us to get there but we shouldn't think of it as the only way of getting there and let's think of this public private school example that we had in a study cafe I won't go through it in detail here just want to remind you this is the study that we're talking about does putting a kid in private school improve their performance and we saw this over here where the claim is does putting kids either in public private school improve the academic achievement and and and it was then clear to us when

we assisted we don't know if the children's ability before they even start school this different so it's actually the children's ability that then decides while clever kids get into private school and and and clever kids also get the higher academic achievement so that's an alternative explanation that it turns out to be the cognitive ability that decides it doesn't really matter which school they go to and in addition we also talked about well maybe it's the parental resources it's it's focusing and having the resources to be able to study at home maybe it's the the values that are being put on education maybe parents that sent the kids to private school just value education much more which then means to end up performing better it's not the school type that makes it perform better it's just because these two things are correlated so again the conclusion could sort of be okay apparently we need a randomized control trial do we need a thousand kids that we randomly put into a public private school and look at them ten years later that would be quite something but if we now collect data on cognitive ability the resources of parents how high education is valued let's say we actually collect some kind of survey data on that then we test the kids ability before they start school we test and question how much parents value education and we have this information and control for this information in the statistical analysis well in the end we could then show if and how school type has an effect on academic achievement we don't have a randomized control trial so so in principle there might be other things that we should have thought of but at least we can in a careful design study where we have data over time it can try to get closer to a conclusion this is really what we want to become better at being sure than a scientific

claim is credible so if we've done all this kind of data collection we can be more sure of our conclusion not necessarily on a person sure but that's all we can do in science is to sort of push our certainty and reduce our uncertainty and that's what we do here this particular example